KINETIC AND POTENTIAL ENERGY WORKSHEET

Name:_____

Determine whether the objects in the following problems have kinetic or potential energy. Then choose the correct formula to use: $E_k = \frac{1}{2} mv^2$ *or* $E_p = mgh$

1. You serve a volley ball with a mass of 2.1 kg. The ball leaves your hand with a speed of 30 m/s. The ball has _____ energy. Calculate it.

   E_k= ? $E_k = \frac{1}{2} m v^2$

   m= 2.1 kg $= \frac{1}{2} \times 2.1 \times 30^2$

   v= 30 m/s Kinetic energy = 945 J

2. A baby carriage is sitting at the top of a hill that is 21 m high. The carriage has mass of 25 kg. The carriage has potential energy. Calculate it.

   E_p= ? $E_p = mgh$

   h= 21 m $= 25 \times 10 \times 21$

   m= 25 kg Potential energy = 5250 J or $\mathbf{5.25 \times 10^3\ J}$

   g= 10 m/s/s

3. What work was done to get the carriage up to the top of the hill in question 2?

   W = E_p all the potential energy is turned into useful work

   Work = $\mathbf{5.25 \times 10^3\ J}$

4. A car is traveling with a velocity of 40 m/s and has a mass of 1120 kg. The car has kinetic energy. Calculate it.

   E_k= ? $E_k = \frac{1}{2} m v^2$

   m= 1120 kg $= \frac{1}{2} \times 1120 \times 40^2$

   v= 40 m/s Kinetic energy = $\mathbf{8.96 \times 10^5\ J}$

5. What work would have to be done by the breaks in order to stop the car in question 4?

   If energy is conserved then:

   W = E_k ∴ Work = $\mathbf{8.96 \times 10^5\ J}$

6. There is a bell at the top of a tower that is 45 m high. The bell weighs 190 N. The bell has E_p energy. Calculate it.

E_p= ? $\quad E_p = mgh$

F_w = 190 N $\quad = F_w \times h$

h= 45 m $\quad = 190 \times 45$

Potential energy $= 8.55 \times 10^3$ J

7. You are on roller blades on top of a small hill. Your potential energy is equal to 1,000.0 Joules. The last time you checked your mass was 60.0 kilograms.

a. What is the height of the hill?

h= ? $\quad E_p = mgh$

E_p= 1000.0 J $\quad 1000 = 60 \times 10 \times h$

m= 60 kg $\quad \frac{1000}{60 \times 10} = h$

g= 10 m/s/s $\quad$ height = 1.67 m

b. If you start skating down this hill, what will be your speed at the bottom of the hill?

v= ? $\quad E_k = \frac{1}{2} m v^2$

E_p= E_k if energy is conserved $\quad 1000 = \frac{1}{2} \times 60 \times v^2$

E_p = 1000.0 J $\quad \frac{2 \times 1000}{60} = v^2$

m= 60 kg $\quad \therefore$ velocity = 5.77 m/s

8. Answer the following:

(a) What is the kinetic energy of a 1 kilogram ball that is thrown into the air with an initial velocity of 30 m/sec?

E_k= ? $\quad E_k = \frac{1}{2} m v^2$

m= 1 kg $\quad = \frac{1}{2} \times 1 \times 30^2$

v= 30 m/s Kinetic energy = 450 J

(b) How much potential energy does the ball have when it reaches the top of its ascent if it doesn't lose any energy?

If energy is conserved then:

E_p= ? $\quad E_p = E_k \therefore$ Potential energy = 450 J

(c) How high into the air did the ball travel?

h= ? $\quad E_p = mgh$

E_p= 450 J $\quad 450 = 1 \times 10 \times h$

m= 1 kg $\quad \frac{450}{1 \times 10} = h$

g= 10 m/s/s $\quad$ height = 45 m